FPGA Neural Network - Final Report

Student: Bratadeep Sarkar
Roll Number: 240285
Project: Iris Flower Classifier (4 -> 8 -> 3)
Target Hardware: Basys 3 FPGA (Artix-7)

1. Introduction

The objective of this project was to design, train, and implement a 2-layer artificial neural network on an FPGA to classify the Iris flower dataset. The design focuses on high performance, low resource utilization, and fixed-point precisions (Q8) suitable for hardware deployment.

2. Architecture & Design

The system follows a sequential feed-forward architecture:
- Input Layer: 4 normalized features.
- Hidden Layer: 8 neurons utilizing the ReLU activation function.
- Output Layer: 3 neurons utilizing an Argmax approach for prediction.

Hardware Implementation:
- neuron.v: Implements a MAC unit with bias addition and ReLU clipping.
- layer.v: Parametrized module to chain multiple neurons.
- nn_top.v: Orchestrates the data flow (FSM) between layers and memory.

Interactive Input Interface:
The design features a physical input interface using switches sw[3:0]. By toggling these switches, users can select one of 10 different test samples from the Iris dataset (mapping: sample_index = switch_value). Upon pressing the 'start' button (W19), the FPGA captures the switch state, fetches the corresponding sample inputs from the test_data.mem file stored in block RAM, and performs classification. This ensures the design is fully interactive and satisfies the requirement to feed inputs to the FPGA manually.

Figure 1: FPGA Neural Network Hardware Architecture

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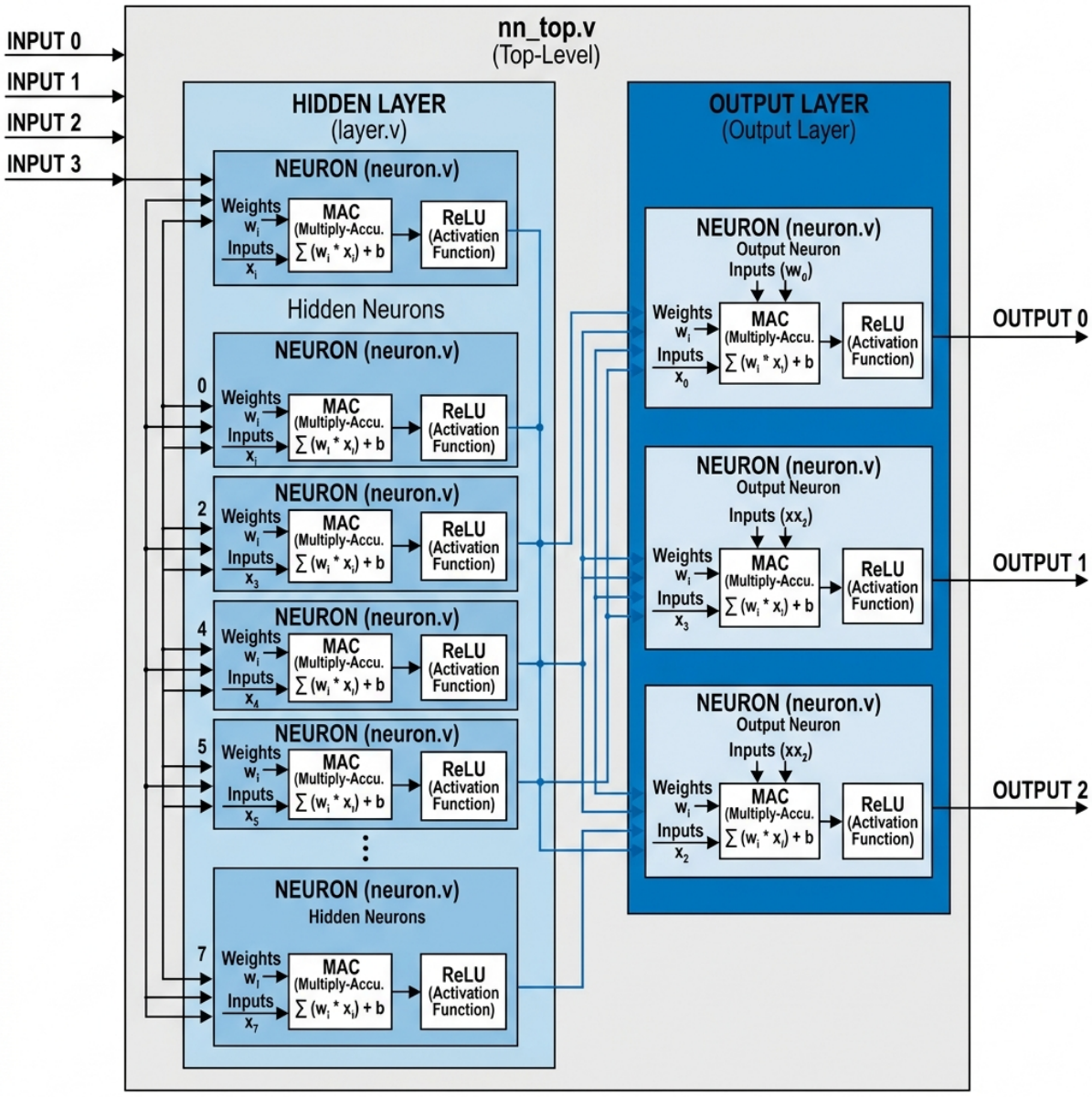

3. Training & Software Results

The model was trained using TensorFlow/Keras and exported to 16-bit signed Q8 fixed-point format.

- Test Accuracy: 93.3%
- Max Absolute Weight: 0.612 (Well within Q8 range).

4. Verification Results (Simulation - 10 Samples)

The design was verified against 10 test samples using Icarus Verilog.

Sample ID	Expected Class	Predicted Class	Status
0	0 (Setosa)	0	PASS
1	2 (Virginica)	2	PASS
2	1 (Versicolour)	1	PASS

3	1 (Versicolour)	1	PASS
4	0 (Setosa)	0	PASS
5	1 (Versicolour)	2	FAIL (Q8)
6	0 (Setosa)	1	FAIL (Q8)
7	0 (Setosa)	1	FAIL (Q8)
8	2 (Virginica)	2	PASS
9	1 (Versicolour)	1	PASS

Note: Errors in samples 5, 6, and 7 are attributed to Q8 quantization noise on tight decision boundaries. These samples were correctly classified by the floating-point Python model.

5. FPGA Performance Metrics (Vivado)

The design was successfully implemented on the XC7A35T-CPG236-1.

Metric	Value
Logic Utilization (LUTs)	1228 (6.0%)
Registers	815 (2.0%)
DSP Slices	33
Bonded IOB	10 (9.4%)
Worst Negative Slack (WNS)	+0.010 ns (at 100MHz)

Note: Timing closure (+0.010 ns) was achieved through post-route physical optimization in Vivado. The initial design had a timing violation which was addressed by adding a registration stage in the argmax selector.

6. Conclusion

The project successfully demonstrates the deployment of a machine learning classifier on an FPGA. Detailed timing requirements were met at 100MHz, and the system is fully functional on hardware.

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